



Food and Agriculture Organization
of the United Nations

Production guidelines for rice



Project: First Season-Long Training of
Trainers on Integrated Rice Crop
Management under the National Rice
Revitalization Programme of the Ministry of
Agriculture (TCP/FIJ/3502)

Technical Cooperation Programme with
the Republic of Fiji

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Rice varieties in Fiji

Improved Varieties.

- MALEKA: Can be grown in poor soil (hill soils), and two crops/year
- UTTAM: Suitable for both dryland/irrigated/wetland and two crops/year
- DEEPAK: Suitable for problem soils (acid sulphate peat) & two crops/year
- NUINUI: Suitable for irrigated & wetland and two crops/year
- TOTOKA: Suited for irrigated/wetland and two crops/year
- STAR: Popular variety and can be grown 3 crops/year under irrigation

Traditional Varieties:

- CHINA MOTKA: Late maturing & low yielding variety, one crop/year
- LAL MOTKA: Late maturing and low yielding, one crop/year

Planting method and seed rate

There are 2 planting seasons for rice: (1) Main Season (November to January); and (2) Off Season (June to August). 3 main methods of planting:

- Broadcasting – use of pre-germinated seeds. Seed @ 30 kg/acre
- Transplanting – seeds are raised in beds & transplanted in main field 3-4 weeks after germination. Seed @ 15 kg/acre. Seedlings are transplanted with 25-30 cm x 25-30 cm spacing with 3 to 4 seedlings per hill.
- Drilling – suitable in dry land using drilling machines. Seed @ 36 kg/acre

Seed sorting/cleaning

- Dissolve 1.5 kg urea/40 liters of water (check solution with an floating egg)
- Soak seeds in urea solution & stir to float unfilled, broken & diseased seeds
- Remove floating poor seeds by hands/sieves
- Wash seeds properly 3-4 times with clean water

Good seed can contribute 10% more yield

Incubation

- Soak a half-filled seed-sack in clean water for 24 hrs, place sack on a wooden platform in a shaded area, cover with rice straw to facilitate incubation

- Turn seeds once after 24 hrs. for aeration and reduce inner-heat of the heap
- Incubate until white "dots" observed
Incubation time: 48 to 72 hrs. (Varies with whether)

Nursery and seeding

- The seed bed is usually prepared 25 to 30 days before transplanting.
- Plough & harrow field 2-3 times until the soil is thoroughly puddled and leveled
- Construct raised beds 10 cm high and 1 – 1.5 m wide with convenient length.
- Construct drainage canals of 40-50 cm between the seed beds
- 300-500 m² seed bed can hold 35–45 kg of seeds with ≥80% germination
- The amount can produce enough seedlings to transplant one hectare of field

Land preparation

- Poor land preparation causes uneven growth and therefore uneven maturity
- Response to fertilizer is also affected by improper land preparation
- Fields need to be well levelled for proper water management.

Wetland Conditions

- Land should be soaked thoroughly with water before ploughing begin.
- Always each ploughing should be followed by 1/2 harrowing
- Fields need to be leveled & puddled before transplanting/broadcasting is done
- For good land preparation, two months period from first ploughing is essential
- For virgin soil under irrigation - proper rotovation & puddling is essential

Dry land Conditions

- Soil should be well tilled, clean trashes of grass, break clots into fine pieces

Fertilizer

- **Urea** – 3 bags¹/ha. To be applied in two splits, half at 3 weeks old, and the remaining half at PI stage for short duration variety but in 3 splits for long duration variety. But for peat areas, urea should be reduced to half.
- **Muriate of Potash** – 2 bags/ha. To be applied as basal (before planting).
- **Single Superphosphate** – 1 bag/ha. To be applied as basal (before planting).

Weed control:

Keep rice field weed free up to maximum tillering stage

- Initial plowing buries weeds, allow germinate their seeds and decomposed by later plowings. Stubble cleaning after every laddering can control weeds.
- Use weed-free seed and keep tools and machinery clean
- Keep levees and irrigation canals, non-farm areas free of weeds
- Weedicides are very effective in weed control:
 - Grasses/broadleaf weeds: spray Stam F34 (Propanil) @ 9-11 lit/ha at 3-4 leaf stage.
 - Broad leaf grasses: spray Rice Spray 70 (MCPA) @ 2-3 lit/ha at 5-6 weeks after sowing
 - Grasses, broadleaf & sedges: Spray Basagran PL 2 (bentazon + propanil) @ 8-10 lit/ha at 3-4 leaf stage
- Avoid rainfall up to 4-8 hours after spraying. Again, Field should be completely drained before application of chemicals and flooded 3 days after spraying
- Mechanical control of weeds by operating finger weeder at 15-20 days after sowing combined with one hand weeding was more economical.

Water management

- Water stress from PI-to early grain filling stage results sterile or under-weight grain
- Excess water after transplanting reduces tillering ability
- Irrigation water should be applied at 2-3 cm depth 3 days after transplanting
- At tillering stage, alternate wetting and drying techniques can be applied

¹ 1 bag = 50 kg

- Water should be drained out from seed plot 7 days before harvest
- Small pond of 2 m depth can facilitate 2/1 irrigation with stored rain water.
- A 15cm high bund can hold rain water and help to face occasional drought in main season.

Pest control

Brown Plant hopper (BPH): The adult hopper is brown colored - both adult & nymph suck sap from the stem. This is a serious pest of Rice in Fiji. The effect on the plants is rapid wilting and in severe outbreaks, the whole field of rice may be destroyed (hopper burn). Need regular monitoring starting from active tillering stage and if 2-4 gravid female or 10 nymph/hill are present in 50% hill – OR 1 hopper/tiller is found - insecticide should be applied.

Control: MIPC (Liquid Form) – 48 mls per 14 lts of water.
 MIPC (Powder Form)- 20gms per 14 lts of water.
 BPMC – 25mls per 14lts of water.
 Spray at 2 to 3 weeks interval at the basal portions of plants.



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BPH sucking sap from rice stem



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Hopper burn

Rice Leaf rollers: Adult moths are light brown in color and wings have dark brown stripes and spots. Larvae are yellow / yellowish green. They lay eggs singly on the leaf surface. The caterpillar of the moth damages the plants by rolling the leaves and feeding on the green tissues. The pest is capable of causing a great reduction in yield if not controlled.



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Adult Moth



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Caterpillar is rolling leaf



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Damaged leaf

Control: If 25% leaves damaged - spray leaves with
Carbaryl – 17gms per 14lts of water.
Dicidex – 28gms per 14lts of water.

Integrated Pest Management (IPM)

IPM involves combinations of control techniques to optimize pest according to local conditions:

- Crop rotation: BPH prone areas vegetables/sugar cane can be inserted
- Keeping field weed free
- Proper spacing: Narrow spacing increases brown plant hopper incidence.
- Perching: to aid insect feeding birds
- Scare birds: can control bird damage
- Light trapping: can attract moths and other insects
- Hand picking: picking of stem borer egg mass
- Plough down stubbles: can control stem borer
- Cultivating legume crops in levee
- Placing straw in levee after harvest to get shelter of beneficial insects
- As last measure with insecticide (Granular, liquid, powder): only when other control measure fails. If applied proper rate & dose must be maintained.

Maturity and post-harvest

- Harvesting can be done when 80-90% grains found mature (straw-color)
- Grain may be threshed by leg on a canvas or using thresher machine
- Clean extra stubbles before drying

- 3-5 days sundry can remove some moisture from grain before storing
- Moisture content of stored grain (food/seed) must be maintained $\leq 12\%$
- Sun-dried grains need to be cool and then stored properly.
- Storing materials should be kept on wooden slab
- Keep different varieties separately.

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